# Nathan Chung

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## TECHNICAL SKILLS

**Cloud:** AWS  Lambda, S3, EC2, EFS, IAM, KMS, CloudFormation, OpenSearch, CodePipeline, CodeBuild
**Infrastructure / DevOps:** Terraform, GitHub Actions, Jenkins, Docker, CI/CD, Nexus
**Languages / Data:** Python, SQL, Bash, Java, JavaScript | Databricks, PySpark, Apache Spark, Apache Airflow

## CERTIFICATIONS

AWS Certified Developer – Associate

## PROFESSIONAL EXPERIENCE

Booz Allen Hamilton  Tysons / Silver Spring, VA | *Apr 2025 – Present*
**Cloud Engineer, U.S. Dept. of Veterans Affairs (VA)**

- Developed and orchestrated AWS infrastructure using CloudFormation and Terraform, improving deployment consistency and reducing manual configuration.
- Designed and deployed AWS Lambda functions via CloudFormation to automate compressed-file processing and IAM access-key rotation.
- Built and managed AWS IAM roles and policies through CloudFormation, supporting secure and compliant access to AWS resources.
- Maintained and optimized 4 AWS CodePipeline CI/CD pipelines across development, staging, and production environments.
- Automated distribution of MuleSoft runtime and license files via Lambda, reducing manual deployment effort and errors.
- Resized and optimized Amazon OpenSearch clusters for cost and performance [1-2K: monthly cost savings].
- Partnered with cross-functional teams to troubleshoot deployment, IAM, and infrastructure issues.

**DevOps Engineer / DXP Developer, Internal Revenue Service (IRS)** | *Jul 2025 – Sep 2025*
*(Concurrent with VA role)*

- Migrated IRS teams' CI/CD pipelines from CloudBees Jenkins to GitHub Actions across .NET, Java, and Lite stacks using standardized templates.
- Built GitHub Actions workflows to package, scan, certify, and publish artifacts to Nexus, enforcing SLSA v1.0 Build Level 3 compliance.

**Data Engineer, Internal** | *Jun 2024 – Mar 2025*

- Developed and deployed ETL/ELT pipelines using Databricks, PySpark, and Python to organize and modernize disparate data sources.
- Collaborated with a multi-disciplinary team of analysts and engineers to support scalable data platforms in an agile environment.

**Cloud Ops Engineer / Systems Administrator, Dept. of Homeland Security (DHS)** | *Jun 2023 – Jun 2024*

- Redesigned AWS server architecture and administered Microsoft Server, RHEL Linux, and AWS EC2 infrastructure, applying security patches to maintain compliance.

**Condé Nast  New York, NY** | *Jun 2022 – Aug 2022*
**Data Science / ML Engineer Intern**

- Automated data validation for Project SPIRE (ML ad-targeting) by building dataset/expectation-suite profilers.
- Designed a backfill workflow that resolved a 6-day data-ingestion lag at batch time.

**Praxis Engineering (General Dynamics)  Annapolis, MD.** | *Apr 2021 – Aug 2021*
**Machine Learning Intern**

- Built a cloud-based video analysis tool (AWS Lambda, EC2, S3) applying YOLOv3 and ResNet50 for object detection and sentiment analysis, reaching 80%+ accuracy.

## EDUCATION

**B.S., Computer Science  University of Maryland, College Park**
**Honors:** University Scholars Program  Science, Discovery, and the Universe (3% acceptance rate)

## PROJECTS

**Unix Shell Reconstruction** | *C, Unix, Emacs*

- Re-engineered a Unix shell to support I/O redirection, command piping, and command chaining via "&&".
- Built a parser to validate user input, execute commands, and apply subshell forking when necessary.

**Stock Price Prediction with Linear Regression** | *Python, scikit-learn*

- Built a linear regression model to predict stock prices 30 days out from adjusted closing price data pulled via Quandl and structured with Pandas.
- Trained and validated the model with an 80/20 train/test split, evaluating prediction accuracy against held-out data.

**WordNet Semantic Graph** | *Ruby*

- Constructed a directed acyclic graph (DAG) representing semantic relationships between words, grouped into synsets with hyponym/hypernym mappings.
- Built a command parser supporting word lookup, least-common-ancestor search, and hyponym-hypernym relationship traversal.

**FIFA Player Data Analysis** | *Python, Jupyter* https://nathanchung20.github.io/

- Ran EDA and built a linear regression model on FIFA player data to identify high-impact skill attributes.